Minimization of Energy Delay Squared Product in Priority Encoder Circuit Design

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EECS312: Digital Integrated Circuits

Prepared for:

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WORST CASE DELAY = 94.68ps
ENERGY * DELAY^2 = 1.077 x 10^8 pJ(ps)^2
SUPPLY VOLTAGE = 1.1 V

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Abstract — **To electrically realize a logical function, the function must first be implemented at the transistor level. Often, this implementation must meet a variety of constraints set by an industry-level specification. Two main constraints to be met are power consumption and propagation delay. To meet these constraints, many optimizations may need to be put in place at several levels of abstraction, from the characteristics of the transistors to the amount and types of gates used in the logic circuit. Our task was to implement a priority encoder, a combinational logic function that takes in n inputs and encodes them into a single $\log_2(n)$ binary output. To achieve this goal, we started at the high level by designing a logic-gate circuit and reducing it through three optimization phases. We then changed various properties at the transistor level to minimize delay and power consumption. To do this, the team used models, properties, and general concepts learned in EECS 312, Digital Integrated Circuits.**

I. Introduction

Our team was tasked by Dr. Dennis Sylvester with creating an 8 to 3 priority encoder. The design for this priority encoder would need to meet a 500pW static power constraint, a 600ps propagation delay constraint, 300ps transition time constraint, and maintain a 10% noise margin. The design would also need to focus on optimizing the Energy*(Delay^2) product. After creating multiple design iterations, we finalized a design incorporating both static CMOS and Dynamic Logic design, as well as multiple power saving tactics.

II. Design Process

The initial design for the priority encoder was built using only static CMOS circuit design. This was done because static CMOS is the most energy efficient design logic, and also the simplest to implement. The first design created used two main logic blocks: a priority resolver and an 8:3 encoder. The priority resolver converted any input to one hot, which was then fed into the encoder. This first design however was significantly above the given constraints for static power and time delay. After considering the circuit, our team realized that the priority resolver logic and the 8:3 encoder logic could be combined and synthesized more efficiently.

The second design created had only one main logic block, which combined both the priority resolver and 8:3 encoder. This design was able to meet the static power constraint or time delay constraint, but was unable to meet both simultaneously. Our largest adjustment was the use of different power level logic gates. For each logic gate (NAND, NOR, Inverter) we had two version, one high static power consumption and one low static power consumption.

The final design included all the above changes, while also adding dynamic logic to only the final inverter to make the time delay constraint. After meeting this constraint, the design parameters were adjusted to make the design as optimized as possible according to the Energy*(Delay^2) product. This optimization, as well as all of the design process, is described in more detail below.

A. Logic Synthesis and Design

$$Q_0 = En \cdot (D_6{}'(D_4 \downarrow D_2)D_1 + D_4{}'D_3 + D_5) + D_7)$$
$$Q_1 = En \cdot (D_4 + D_5 + D_2 \downarrow D_3) \uparrow (D_6 \downarrow D_7)$$
$$Q_2 = En \cdot (D_4 \downarrow D_5) \uparrow (D_6 \downarrow D_7)$$

Fig. 1. Final boolean logic equations combining the priority resolver and encoder and favoring NAND/NOR gates.

The first step to designing the priority encoder was to determine the boolean expression for its functionality and reduce it. By separating it into a priority resolver and an 8-3 encoder, then taking the minterm (sum of products) of their truth tables, an expanded boolean expression was found for each output, Q_0, Q_1, and Q_2. Using these expressions, two logic gate-level circuits were synthesized: one for the priority resolver and one for the encoder. By connecting the output of the priority resolver to the input of the encoder, the full priority encoder was synthesized.

However, due to a need for lower delay and power consumption, the two circuits were then combined and reduced as a single circuit. After combining the boolean expressions for each, the overall expressions for the outputs were reduced greatly using boolean algebra, leading to the fully reduced boolean expressions for an 8-3 priority encoder. As an additional optimization, the expressions were manipulated in a third reduction phase, sacrificing the simplicity of the boolean logic to make use of as many NAND and NOR gates as possible. These are the equations that are shown in Figure 1.

Despite increasing the number of operations in the boolean expression, placing priority on these gates decreased the number of overall transistors in the logic circuit, due to NAND and NOR gates requiring fewer transistors to create than AND

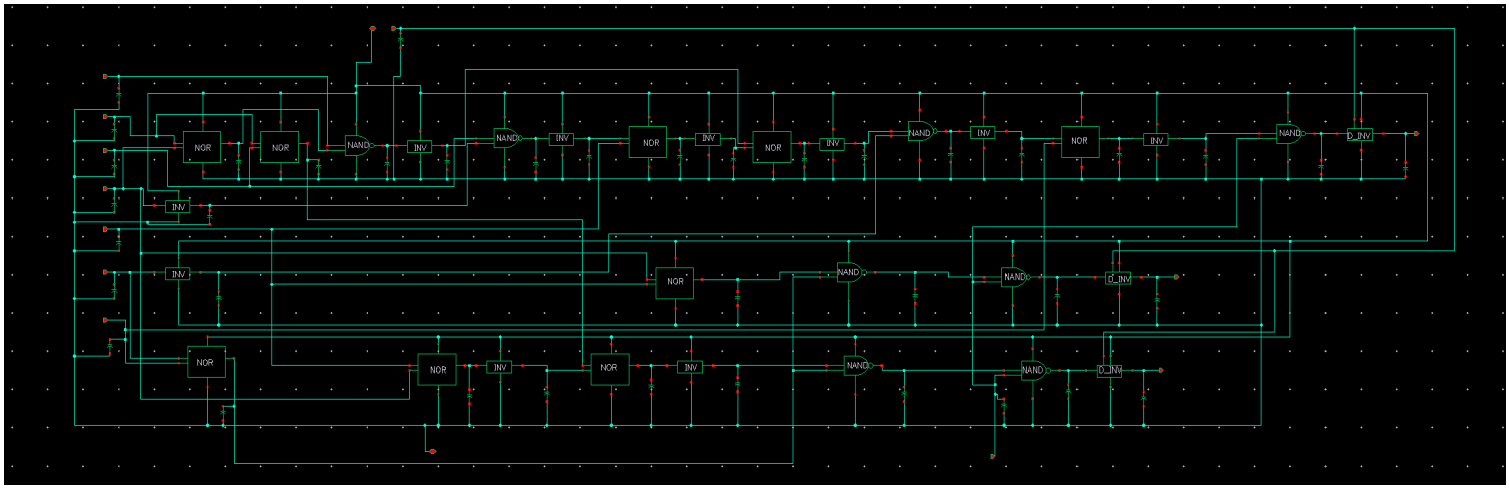

Fig. 2. Full Schematic of Priority Encoder

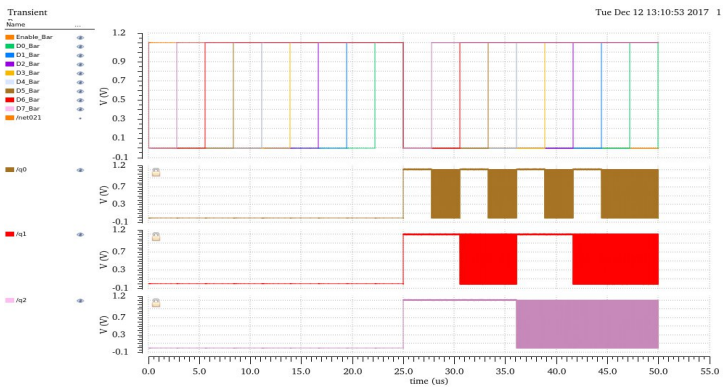

Fig. 3. Input Deck Functionality Test

		Width	Length	Vth
NOR	PMOS	240nm	60nm	0mV
	NMOS	120nm	60nm	+200mV
NAND	PMOS	120nm	60nm	-200mV
	NMOS	120nm	60nm	+200mV
INV	PMOS	240nm	60nm	-100mV
	NMOS	120nm	60nm	+100mV
D_INV	PMOS	1.57um	60nm	0mV
	NMOS	930nm	60nm	0mV

Fig. 4. Gate Parameters

and OR gates. After this phase, the logic synthesis and design of the priority encoder was complete, and thus the team moved onto other methods of optimization. A full schematic can be found in Figure 2. The full simulation of the input deck functionality check is in Figure 3. The gate parameters for the schematic can be found in Figure 4.

B. Time Delay

The worst-case propagation delay constraint was given as 600ps. This constraint was to only be applied to transitions from 0 to 1 on the outputs of the priority encoder. Additionally, a constraint of 300ps for rise and fall time on the outputs was to be enforced. Before adjusting any transistor characteristics, our first goal was to create a static CMOS circuit that exhibited the correct functionality.

In our early static CMOS circuit, the logic was not reduced, leading to many gates. Verification and delay testing revealed the circuit's delay to be approximately 3 orders of magnitude greater than the constraint of 600ps. The group decided that this delay was much too great to fix simply by tweaking individual transistor characteristics. Therefore, we decided to try to further simplify the logic circuit and implement dynamic counterparts to the gates we were currently using.

We decided on boolean expressions that focused on the use of NOR and NAND gates rather than OR and AND gates, as it eliminated the need to attach an inverter to many of our gates. At this point, we decided to change the individual paths to use a dynamic inverter at each output, rather than a static CMOS inverter. This decreased our delay significantly by adding a precharge phase for our circuit

$$t_{pLH} = 0.69\frac{3}{4}\frac{C_L V_{dd}}{I_{dsatp}} = 0.52\frac{C_L V_{dd}}{\frac{W_P}{L}k_p' V_{dsatp}(V_{dd}-V_{thp}-\frac{V_{dsatp}}{2})}$$

$$600 * 10^{-12} = 0.52\frac{(50*10^{-15})(1.1)}{\frac{W_P}{(60*10^{-9})}(118*10^{-6})(0.488)(1.1-.412-\frac{0.488}{2})} \rightarrow W_p = 112 \text{ nm}$$

Fig. 5. Calculations to determine minimum PMOS width for propagation delay

The last change we made to decrease delay was increasing the size of several of our output inverter transistors. This also played a main role in pushing the circuit to meet the rise/fall time constraint of 300ps. It is worth noting that while the theoretical requirement for the PMOS transistor in these output inverters was calculated to be approximately 112 nm to meet the delay requirement, seen in Figure 5. However, to meet the rise time requirements, we increased the size to 1.2um. This had consequences relating to our power consumption, which had to be accounted for with other optimizations (discussed in Static Power and Energy sections). After optimizing the Wp to 1.57um, the worst case delay was calculated as 94.68ps, as seen in Figure 6 below.

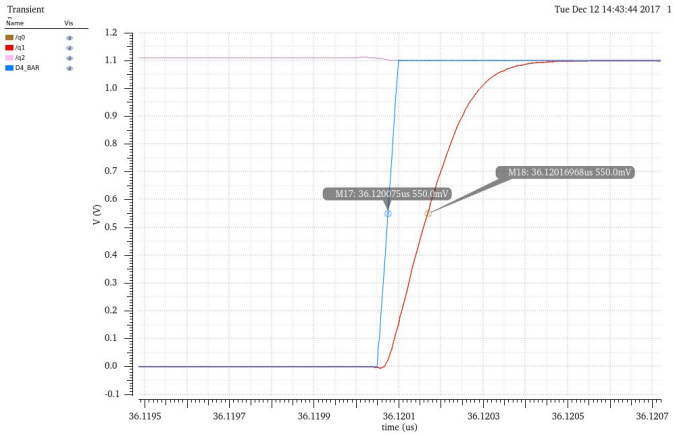

Fig. 6. Worst Case Delay

All the dynamic inverters at the output nodes of the encoder are sized the same resulting in equivalent critical paths for the 0→1 transitions for Q_0, Q_1, and Q_2. This path is from the clock driver to the dynamic inverter to the output node.

C. Static Power

The static power constraint given was 500pW. To reduce the leakage current in our logic gates, we created low static power consumption logic gates by increasing the threshold voltage of CMOS gate transistors. These low static power gates were much slower, so they were added to the non-critical paths to decrease static power consumption without increasing worst case delay. We also changed Vdd to 1.1V to help reduce static power consumption. This put our static power below the 500pW constraint for static CMOS, but our delay increased above the constraint of 600ps.

In the next design iteration when Dynamic Logic was added, static power and time delay decreased to the point where all the non-dynamic gates were changed to our low power gates. The low power gates were used because it would only affect the delay of 1→0 transitions which were not under constraint. This also would not affect our 0→1 transitions which were only dependent on dynamic gates. At this point in the design an inverter chain was attempted to drive output, but the static power consumption was too high. The only parameter that we found to have significant change in static power in the stage driving the output load was the Wp and Wn of the final dynamic inverter.

Using the minimum Wp from Time Delay did not work because the inverter was not strong enough to drive the 50fF load capacitance within the rise time and fall time constraints. The minimum Wn size was chosen to barely meet the fall time constraint in order to minimize subthreshold leakage.

We then used the largest transistor value (3.11um) from the inverter chain as a starting value for a single driving inverter. This value however optimizes for delay, while we were still concerned with meeting the static power consumption and transition time constraints by the minimum amount possible.

Our delay was low enough that we were willing to increase it to meet the static power constraint. We also took advantage of how static power was calculate from the average of the given inputs, and made it so the average was as close as possible to the 500pW constraint despite some input combinations being above 500pW.

We used a parametric analysis of Wp and Wn and found the optimal widths to be Wp = 1.57um and Wn = 930nm. With these widths, and all the above design parameters, static power consumption was 499.29pW, with rise time equaling 197.9ps and fall time equaling 286.7ps.

The project specifications state that when ENABLE is low, then CLK is high. However, it does not specify what CLK should be at when ENABLE is high. As directed by our GSI Mr. Pradeep Chandramouli we chose the CLK voltage which was most beneficial. For us, this turned out to be that ENABLE and CLK would always be opposite.

D. Noise Margins

The noise margins must exceed 10% of Vdd. The final inverter in our design is a dynamic inverter, and the sizing on that inverter will determine whether the output will be rail to rail and meet noise margin requirements. Analysis was done for the chosen sizings on the dynamic inverter that met the static power and delay constraints. Based off this analysis the chosen widths met the noise margin constraints.

E. Energy

Despite Static CMOS being the most energy efficient design, it was not possible to have both the delay and static power constraints met given our logic design. At first, we only changed our critical path to dynamic logic, which did decrease the delay by a significant factor but at the cost of a large increase in power. Then we switched only the final inverter in the critical path to dynamic to reduce dynamic power consumption while keeping a low delay.

The critical path delay for Q_0 then decreased from 680ps to 94.68ps, a reduction by a factor of 7.18. Our new worst delay then became approximately 480ps for both Q_1 and Q_2 Given that we needed to use one dynamic gate, we then decided to transfer all final output inverters to dynamic. This increased the total power by a factor of 3, but worst case delay was then reduced by a factor of 5.07. Using the $energy*delay^2$ goal this resulted in a decrease by a factor of approximately 8.57.

To drive the load capacitance while staying under rise and fall time constraints, we needed to select a Wp for the dynamic inverters to barely meet the static power constraint while making the 0→1 delay as fast as possible. We found that the maximum Wp to meet static power was 1.57um. This resulted in a rise time of 197.9ps, a delay of 94.68ps, and an average static power consumption of 499.29pW.

All of these changes resulted in an energy consumption of 12.01nJ.

III. Results

Using all the above changes, the table below shows all the circuit characteristics analyzed in this project.

TABLE 2
CIRCUIT CHARACTERISTICS

Characteristic	Measurement
Worst Case Delay	94.68ps
Static Power Consumption	499.29pW
Total Energy (1ms)	12.01nJ
Worst Case Rise Time	197.9ps
Worst Case Fall Time	286.7ps
Energy * Delay^2	1.077 x 10^8 pJ(ps^2)

IV. Team Dynamic

As a group of three, we spent the early stages of the project evaluating the group members' skills and agreeing on work delegations accordingly. Based on experience with logic synthesis, Alex Kisil was delegated the task of designing and reducing the boolean expressions for the priority encoder's outputs, as well as translating them to a full logic circuit. As Sanjee Choudhuri was proficient in Cadence Virtuoso, he was delegated the task of creating the design blocks to be used in our circuit and creating the overall circuit. This included inverters, NAND gates, and NOR gates amongst others. Luke Heberling played the role of using models and running simulations. Together we all decided what characteristics needed to be improved upon, and theorized methods of how to do so.

Beyond this point, all members of the team were responsible for coming up with creative ideas to decrease power and delay in their respective parts of the circuit, as well as giving additional perspectives on other teammates' ideas. Every potential design change went through a process of being discussed and criticized between all three members of the group before being implemented. Furthermore, every new change was followed by a comprehensive test for how it affected each characteristic of the circuit. No changes were made to the circuit without all three members being involved.

V. Conclusion

The group was tasked with exploring different methods of gate and transistor-level design to implement a priority encoder. A large part of this assignment was to meet specific constraints on propagation delay, transition times, noise margins, and static power consumption. To meet these constraints, we utilized various methods such as minimizing gate-level logic through boolean algebra, switching gates to dynamic logic to decrease delay, altering the size of some transistors to adjust transition times, and fine-tuning gate parameters to minimize power.

A large factor in decreasing delay and power consumption was reducing the logic circuit to utilize the fewest transistors possible. The boolean expressions for the priority encoder's outputs went through three phases of simplification. They started at the most expanded form of the functions minterms, then were finalized as the most compressed forms while still preferring NANDs and NORs to reduce the total amount of transistors implemented. Using less transistors decreased both delay and power consumption.

The early version of our circuit only made use of static CMOS logic. By switching the inverters at the end of each output's path to dynamic logic, we were able to decrease delay greatly. This also reduced static power consumption, but increased dynamic power significantly. To compensate for this increase in power, we switched most of our static CMOS gates to utilize high-threshold voltage transistors which decreased leakage current.

Furthermore, to decrease transition times, the widths of the dynamic inverters' transistors were increased. This made both the pull-up and pull-down networks stronger, so the output would be able to transition more quickly. However, in order to balance this decrease in transition with the increase in power that it caused, we had to carefully adjust the widths according to a parametric sweep run on the widths' impacts. Bringing the NMOS width to 930nm and the PMOS width to 1.57um resulted in a maximum rise time of 197.9ps and a maximum fall time of 286.7ps.

Noise margins were closely monitored and maintained throughout the production of the project, and did not pose any noteworthy challenges when optimizing the circuit.

Through these methods we were able to meet all required constraints and ultimately bring our final energy delay-squared product to 1.077 x 10^8 pJ(ps)^2.